

Module 5: Program Listing

5.1 Cost Estimation

User Inputs:

1. User Registration & Profile Setup
2. Login (Google OAuth / Email)
3. Text Input for Handwriting Synthesis (up to 1600 characters)
4. Style Selection (0-12 handwriting styles)
5. Bias/Neatness Adjustment (0.0-1.5 range)
6. Stroke Color Selection (color picker)
7. Stroke Width Adjustment (1-5 pixels)
8. Paper Type Selection (white, cream, aged, lined, grid, recycled)
9. Ink Type Selection (ballpoint, gel, fountain, marker, pencil)
10. Wear Level Adjustment (0.0-1.0)
11. Batch Generation Configuration (multiple styles)
12. Gallery Filtering & Search
13. Favorite/Tag Management

User Outputs:

1. Generated Handwriting SVG Display
2. Realistic PNG Rendering Preview
3. Gallery Grid View with Thumbnails
4. Download Options (SVG/PNG/PDF)
5. Credit Balance Display
6. Generation Status Indicators
7. Batch Job Progress Tracking
8. Error Messages & Validation Feedback

User Enquiries:

1. Search Gallery (by text, tags, date)
2. Filter by Style/Status
3. View Generation Details
4. Check Credit Balance

Number of Files (Internal Logical Files):

1. Users
2. SavedGenerations
3. BatchJobs
4. SynthesisUsage
5. Accounts (OAuth)

6. Sessions

Number of External Interfaces:

1. Authentication Provider (Google OAuth 2.0)
2. Handwriting Synthesis API (Python FastAPI)
3. File Storage Provider (UploadThing)
4. Background Job Processor (Inngest)

Domain Characteristic Table

MEASUREMENT PARAMETER	COUNT (value >= 0)	WEIGHTING FACTOR
		Simple
Number of User Inputs	13	
Number of User Outputs	8	
Number of User Inquiries	4	
Number of Files	6	
Number of External Interfaces	4	

Complexity Adjustment Table

ITEM	COMPLEXITY ADJUSTMENT QUESTIONS	SCALE (0-5)
1	Does the system require reliable backup and recovery?	4
2	Are data communications required?	5
3	Are there distributed processing functions?	5
4	Is performance critical?	4
5	Will the system run in an existing, heavily utilized operational environment?	3
6	Does the system require on-line data entry?	5
7	Does the on-line data entry require the input transaction to be built over multiple screens or operations?	4
8	Are the master files updated on-line?	5
9	Are the inputs, outputs, files or inquiries complex?	4
10	Is the internal processing complex?	5
11	Is the code to be designed reusable?	5
12	Are conversion and installation included in the design?	4
13	Is the system designed for multiple installations in different organizations?	3

14	Is the application designed to facilitate change and ease of use by the user?	5
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Total Complexity Adjustment Factor (CAF): 61

Function Point Calculation

Unadjusted Function Points (UFP):

Parameter	Count	Simple	Average	Complex	Total
User Inputs	13	-	-	13 x 6	78
User Outputs	8	-	8 x 5	-	40
User Inquiries	4	-	-	4 x 6	24
Files	6	-	6 x 10	-	60
External Interfaces	4	-	-	4 x 10	40
UFP Total					242

Value Adjustment Factor (VAF): $VAF = 0.65 + (0.01 \times CAF) = 0.65 + (0.01 \times 61) = 1.26$

Adjusted Function Points (AFP): $AFP = UFP \times VAF = 242 \times 1.26 = 305.0$

RESULT

PROJECT FUNCTION POINTS 305.0

5.2 Schema Design

Database Tables Overview

The application uses PostgreSQL with Prisma ORM. The schema consists of the following main tables:

- `_prisma_migrations`
 - `accounts`
 - `users`
 - `saved_generations`
 - `batch_jobs`
 - `synthesis_usage`
 - `sessions`
 - `verification_tokens`
-

Users Table

Column	Type	Constraints	Description
id	VARCHAR(25)	PRIMARY KEY	CUID identifier
name	VARCHAR(255)	NULLABLE	Display name
email	VARCHAR(255)	UNIQUE, NULLABLE	Email address
emailVerified	TIMESTAMP	NULLABLE	Verification date
image	VARCHAR(500)	NULLABLE	Profile image URL
password	VARCHAR(255)	NULLABLE	Bcrypt hashed password
credits	INTEGER	DEFAULT 10	Available synthesis credits
createdAt	TIMESTAMP	DEFAULT NOW()	Account creation
updatedAt	TIMESTAMP	ON UPDATE	Last modification

SavedGenerations Table

Column	Type	Constraints	Description
id	VARCHAR(25)	PRIMARY KEY	CUID identifier
userId	VARCHAR(25)	FOREIGN KEY -> Users	Owner reference
status	ENUM	DEFAULT 'COMPLETED'	PENDING, GENERATING, COMPLETED, FAILED
text	TEXT	NOT NULL	Input text content
style	INTEGER	NOT NULL	Style index (0-12)
bias	FLOAT	NOT NULL	Neatness value (0.0-1.5)
strokeColor	VARCHAR(50)	DEFAULT 'black'	CSS color value
strokeWidth	INTEGER	DEFAULT 2	Stroke width (1-5)
fileUrl	VARCHAR(500)	NULLABLE	UploadThing URL
fileKey	VARCHAR(100)	UNIQUE, NULLABLE	UploadThing file key
svgContent	TEXT	NULLABLE	Raw SVG markup
realisticPng	TEXT	NULLABLE	Base64 encoded PNG
paperType	VARCHAR(20)	NULLABLE	Paper type identifier
inkType	VARCHAR(20)	NULLABLE	Ink type identifier
wearLevel	FLOAT	NULLABLE	Degradation (0.0-1.0)

isFavorite	BOOLEAN	DEFAULT FALSE	Favorite flag
tags	TEXT[]	DEFAULT []	Array of tag strings
batchJobId	VARCHAR(25)	FOREIGN KEY -> BatchJobs	Batch reference
createdAt	TIMESTAMP	DEFAULT NOW()	Creation timestamp

Indexes: userId, status, batchJobId

BatchJobs Table

Column	Type	Constraints	Description
id	VARCHAR(25)	PRIMARY KEY	CUID identifier
userId	VARCHAR(25)	FOREIGN KEY -> Users	Owner reference
name	VARCHAR(255)	NULLABLE	Job display name
text	TEXT	NOT NULL	Source text
totalVariants	INTEGER	NOT NULL	Number of variants
creditsUsed	INTEGER	DEFAULT 0	Credits consumed
status	ENUM	DEFAULT 'PENDING'	PENDING, PROCESSING, COMPLETED, FAILED
completedCount	INTEGER	DEFAULT 0	Completed count
createdAt	TIMESTAMP	DEFAULT NOW()	Creation timestamp

Indexes: userId, status

SynthesisUsage Table

Column	Type	Constraints	Description
id	VARCHAR(25)	PRIMARY KEY	CUID identifier
userId	VARCHAR(25)	FOREIGN KEY -> Users	User reference
creditsUsed	INTEGER	NOT NULL	Credits consumed
linesCount	INTEGER	NOT NULL	Lines generated
charactersCount	INTEGER	NOT NULL	Characters processed
style	INTEGER	NOT NULL	Style used

bias	FLOAT	NOT NULL	Bias value
success	BOOLEAN	DEFAULT TRUE	Success status
createdAt	TIMESTAMP	DEFAULT NOW()	Usage timestamp

Indexes: userId, createdAt

Accounts Table (OAuth)

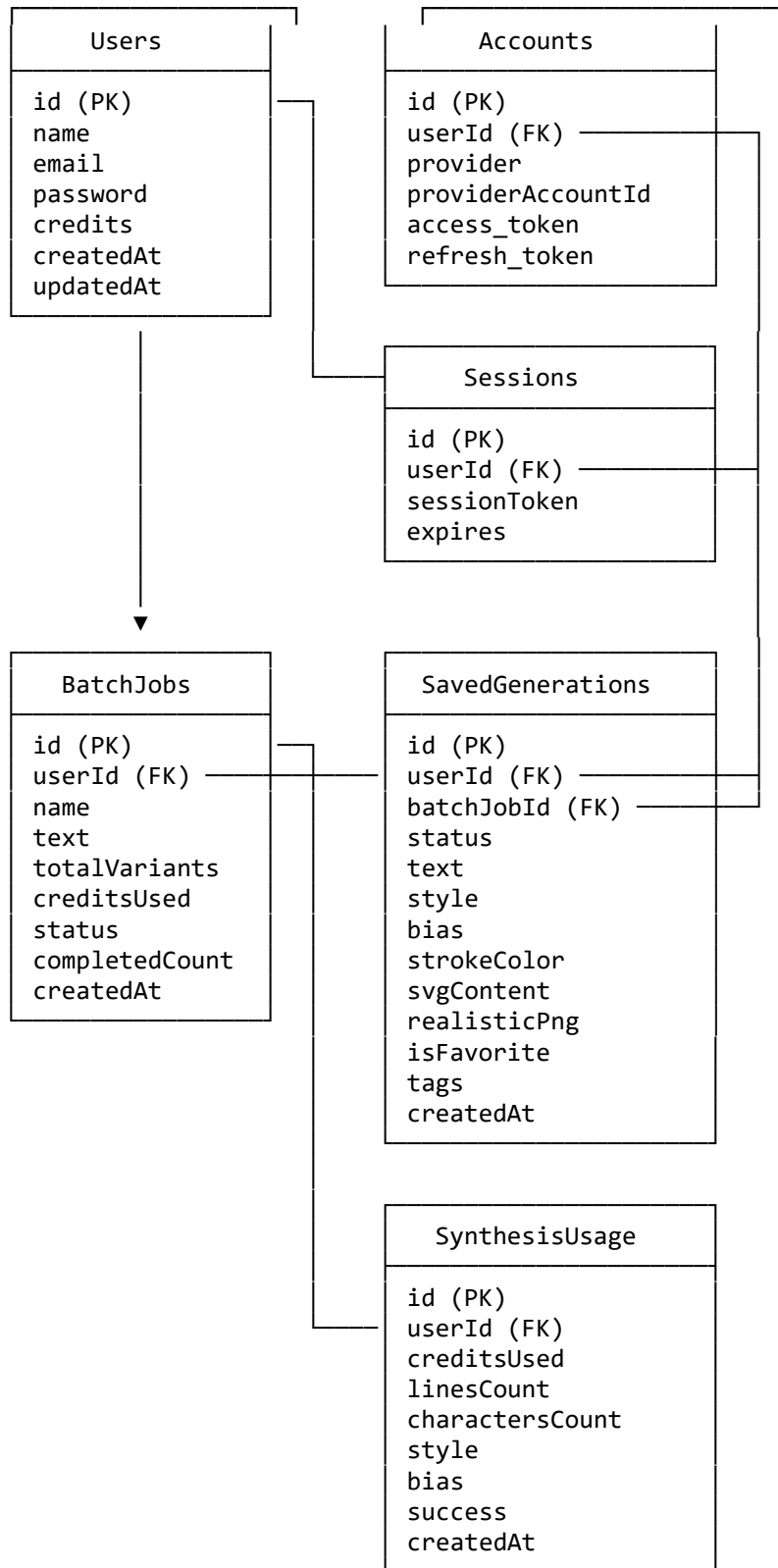
Column	Type	Constraints	Description
id	VARCHAR(25)	PRIMARY KEY	CUID identifier
userId	VARCHAR(25)	FOREIGN KEY -> Users	User reference
type	VARCHAR(50)	NOT NULL	Account type
provider	VARCHAR(50)	NOT NULL	OAuth provider
providerAccountId	VARCHAR(255)	NOT NULL	Provider's user ID
access_token	TEXT	NULLABLE	OAuth access token
refresh_token	TEXT	NULLABLE	OAuth refresh token
expires_at	INTEGER	NULLABLE	Token expiration
token_type	VARCHAR(50)	NULLABLE	Token type
scope	VARCHAR(255)	NULLABLE	OAuth scopes
id_token	TEXT	NULLABLE	ID token

Unique Constraint: (provider, providerAccountId)

Sessions Table

Column	Type	Constraints	Description
id	VARCHAR(25)	PRIMARY KEY	CUID identifier
sessionToken	VARCHAR(255)	UNIQUE	Session token
userId	VARCHAR(25)	FOREIGN KEY -> Users	User reference
expires	TIMESTAMP	NOT NULL	Expiration time

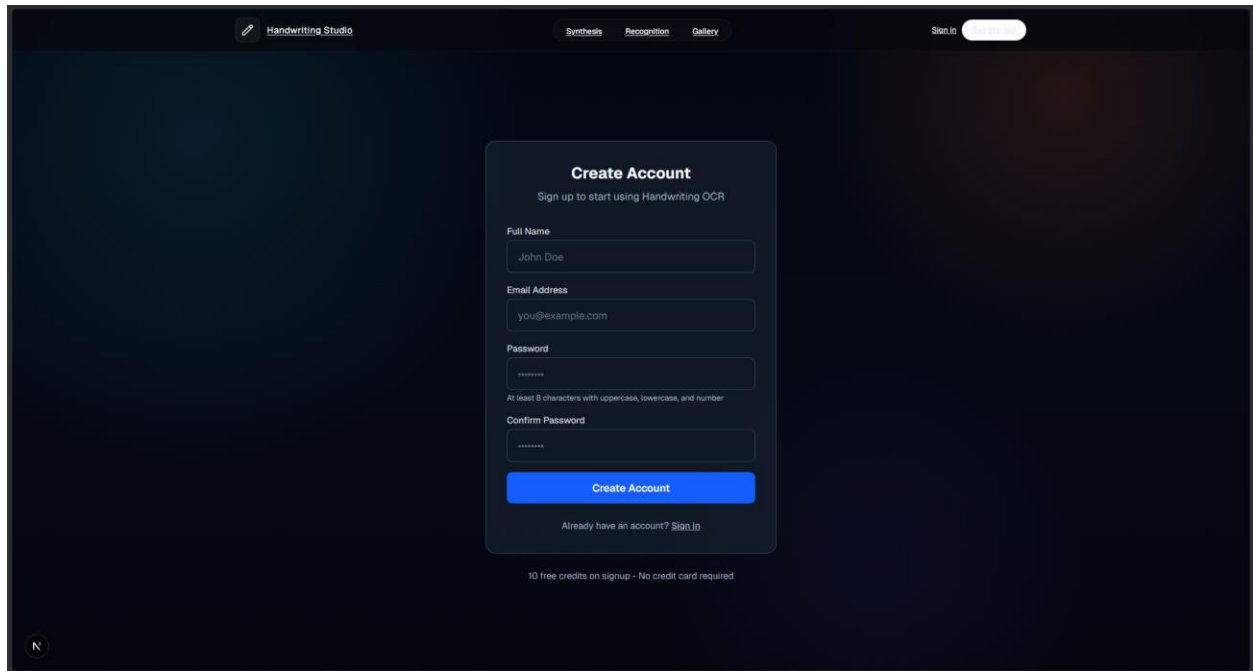
Entity Relationship Diagram



5.3 User Manual With Screenshots

5.3.1 Getting Started

Registration: 1. Navigate to the application URL 2. Click “Sign Up” button 3. Enter your full name, email, and password 4. Alternatively, click “Continue with Google” for OAuth login 5. Upon successful registration, you receive 10 free credits



The screenshot shows the 'Create Account' form for 'Handwriting Studio'. The form is centered on a dark background. At the top, there's a navigation bar with 'Handwriting Studio' on the left and 'Sign In' and 'Get Started' on the right. The main form has the title 'Create Account' and a subtitle 'Sign up to start using Handwriting OCR'. It contains four input fields: 'Full Name' (with 'John Doe' entered), 'Email Address' (with 'you@example.com' entered), 'Password' (with a strength indicator), and 'Confirm Password'. Below these fields is a blue 'Create Account' button. At the bottom of the form, there's a link 'Already have an account? Sign In'. A footer note at the very bottom states '10 free credits on signup - No credit card required'.

Handwriting Studio

Synthesis Recognition Gallery

Sign In Get Started

Create Account

Sign up to start using Handwriting OCR

Full Name

John Doe

Email Address

you@example.com

Password

At least 8 characters with uppercase, lowercase, and number

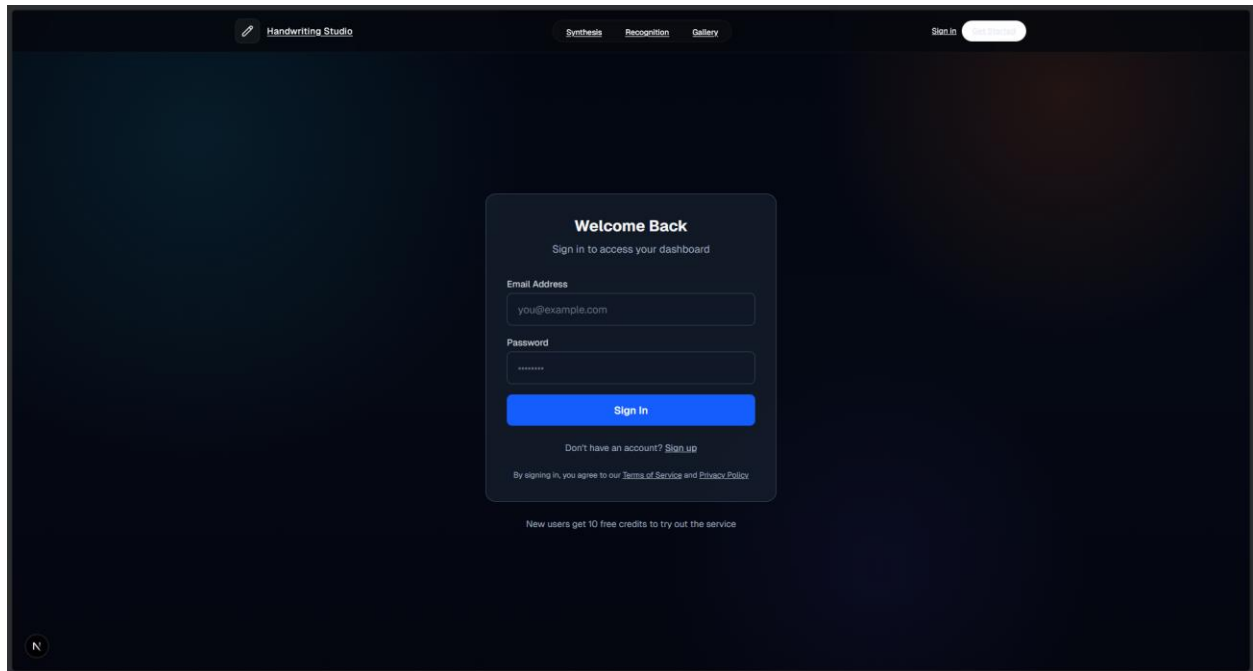
Confirm Password

Create Account

Already have an account? [Sign In](#)

10 free credits on signup - No credit card required

Login: 1. Navigate to the login page 2. Enter your email and password 3. Click “Sign In” or use Google OAuth 4. You will be redirected to the dashboard



The image shows a dark-themed login page for 'Handwriting Studio'. At the top, there is a navigation bar with the logo 'Handwriting Studio' on the left, and links for 'Synthesis', 'Recognition', 'Gallery', and 'Sign In' on the right. The 'Sign In' link is highlighted with a white pill-shaped background. In the center of the page is a light gray rounded rectangle containing the login form. The form has a heading 'Welcome Back' followed by the text 'Sign in to access your dashboard'. Below this are two input fields: 'Email Address' with the placeholder 'you@example.com' and 'Password' with a masked password 'password'. A blue 'Sign In' button is positioned below the password field. Under the button, there is a link 'Don't have an account? Sign up' and a line of text stating 'By signing in, you agree to our [Terms of Service](#) and [Privacy Policy](#)'. At the bottom of the form, a note says 'New users get 10 free credits to try out the service'. In the bottom left corner of the page, there is a small circular icon with the letter 'N'.

Handwriting Studio

Synthesis Recognition Gallery Sign In

Welcome Back

Sign in to access your dashboard

Email Address

you@example.com

Password

password

Sign In

Don't have an account? [Sign up](#)

By signing in, you agree to our [Terms of Service](#) and [Privacy Policy](#)

New users get 10 free credits to try out the service

N

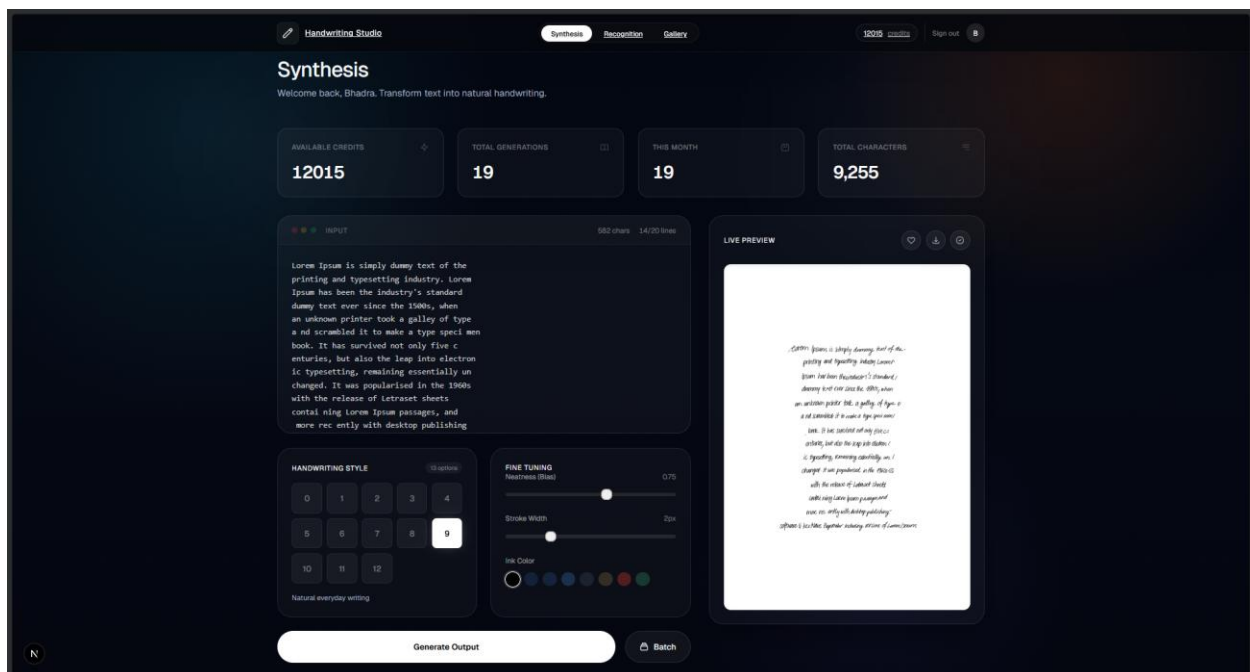
5.3.2 Generating Handwriting

Step 1: Enter Text - Navigate to the “Synthesis” tab on the dashboard - Enter your text in the text area (maximum 1600 characters) - Text supports multiple lines (up to 20 lines, 75 characters per line) - Supported characters: a-z, A-Z (except Q, X, Z), 0-9, common punctuation

Step 2: Configure Style Options - **Style Selection:** Choose from 13 unique handwriting styles (0-12) - **Bias/Neatness:** Adjust the slider (0.0 = messy, 1.5 = very neat) - **Stroke Color:** Click the color picker to select ink color - **Stroke Width:** Adjust stroke thickness (1-5 pixels)

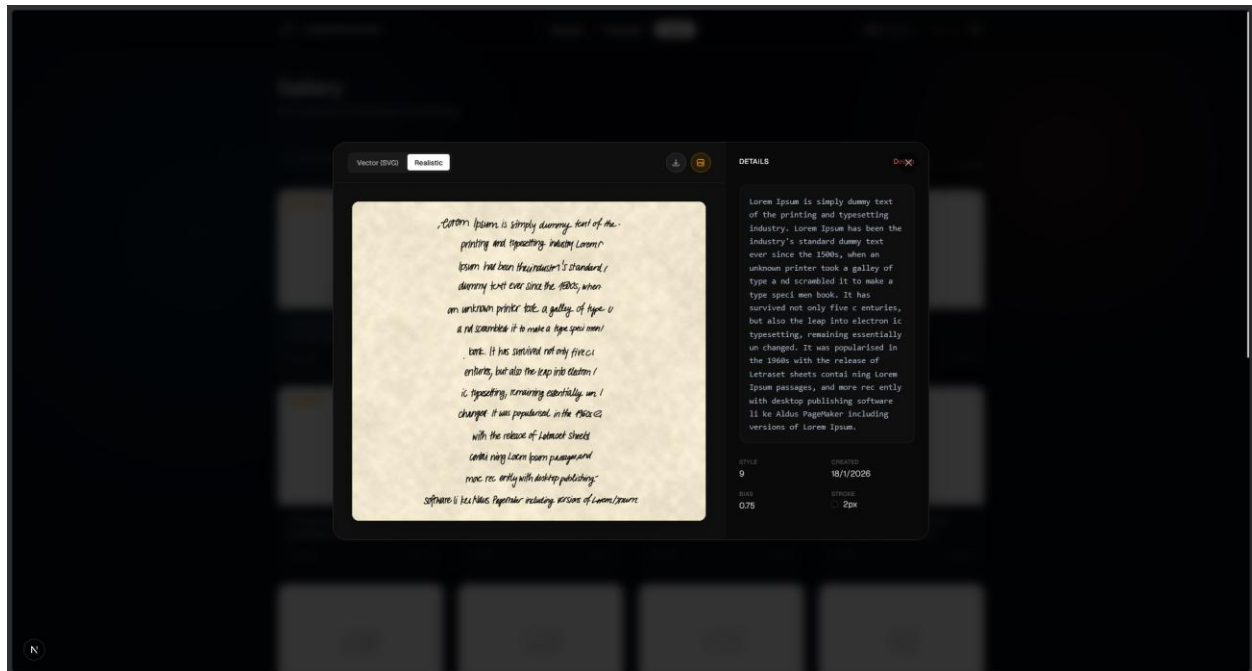
Step 3: Generate - Click the “Generate Handwriting” button - Wait for the synthesis to complete (typically 2-5 seconds) - The generated handwriting SVG will appear in the preview area - 1 credit is deducted per generation

Step 4: Save or Download - Click “Save to Gallery” to store the generation - Click “Download SVG” to download the vector file - Click “Download PNG” to download as an image



5.3.3 Realistic Rendering

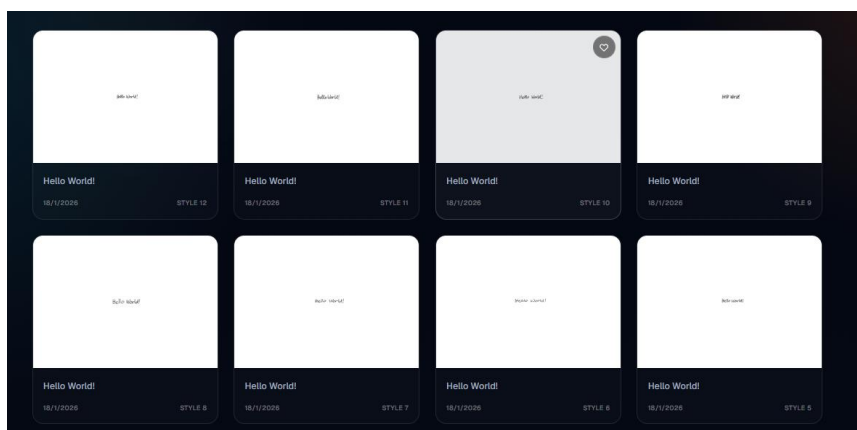
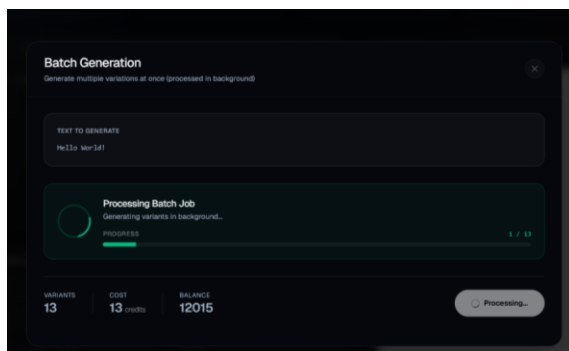
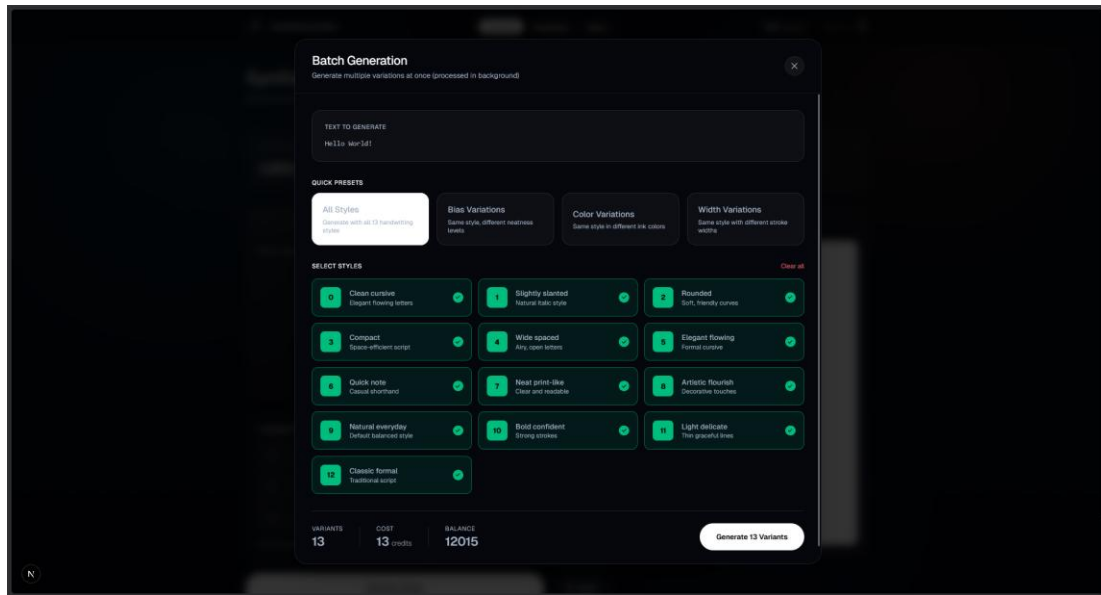
Make it Look Handwritten: 1. From the gallery, select a saved generation 2. Click “Make Realistic” button 3. Configure realistic rendering options: - **Paper Type:** white, cream, aged, lined, grid, recycled - **Ink Type:** ballpoint, gel, fountain, marker, pencil - **Wear Level:** 0.0 (pristine) to 1.0 (heavily worn) 4. Click “Apply” to generate the realistic PNG 5. Download the realistic version



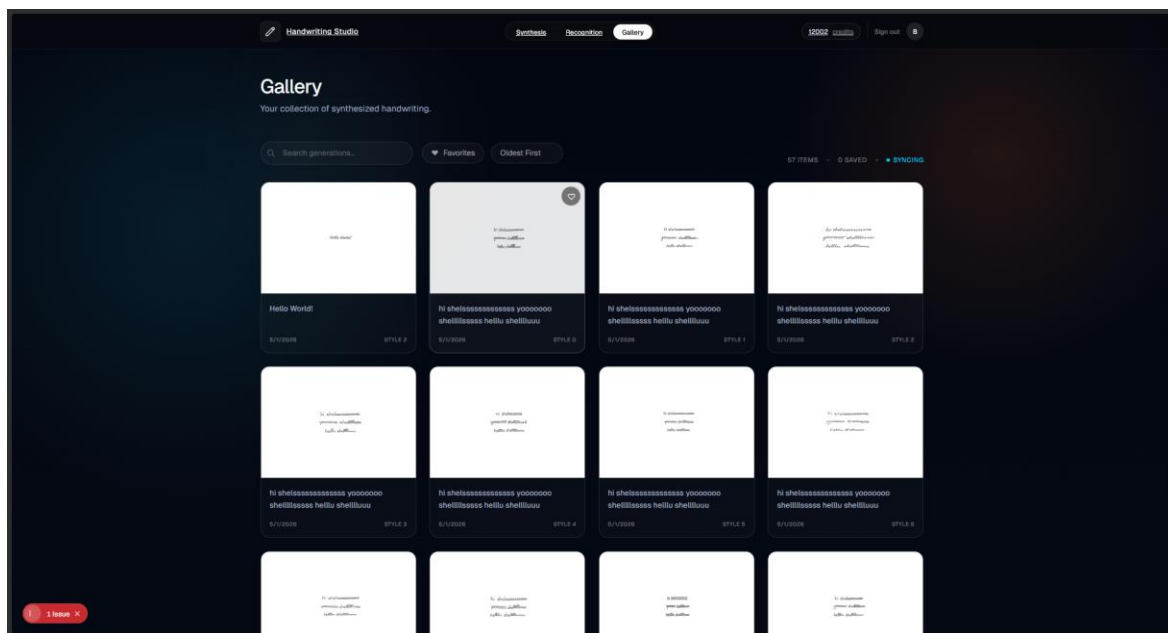
5.3.4 Batch Generation

Generate Multiple Variants: 1. Navigate to “Batch” tab 2. Enter your text content 3. Select multiple styles you want to generate 4. Configure other parameters (bias, colors) 5. Click “Generate Batch” 6. Monitor progress in the batch jobs panel 7. View all variants in the gallery when complete

Note: Batch generation requires sufficient credits for all selected variants.



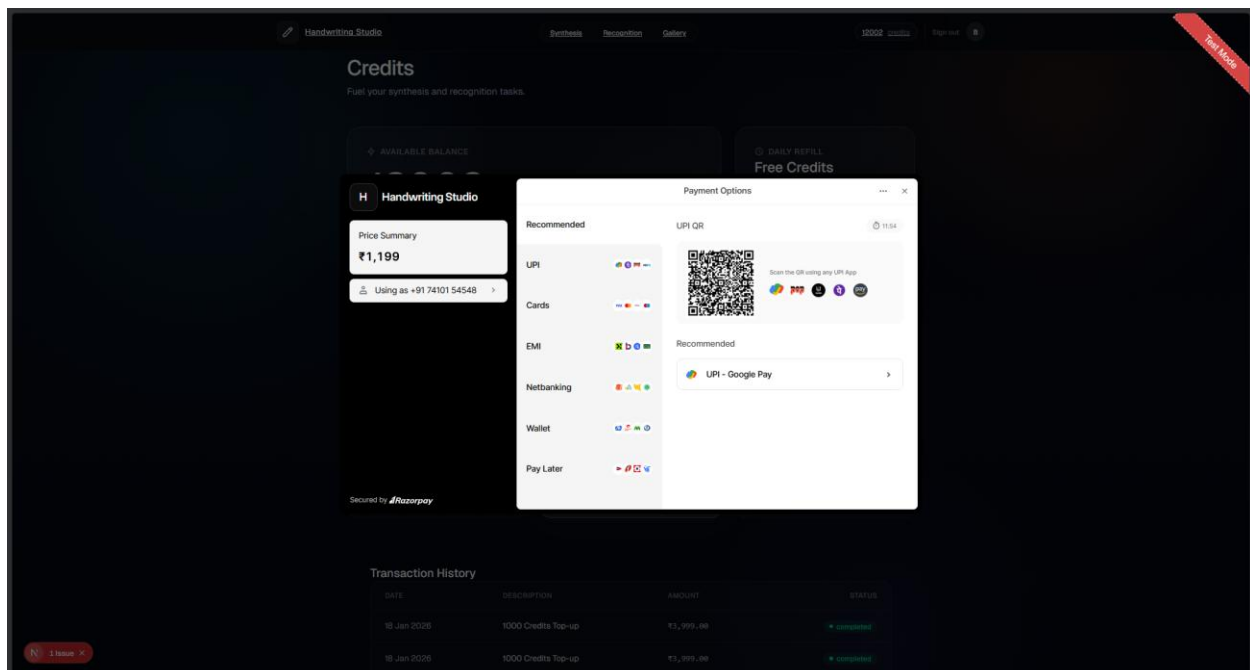
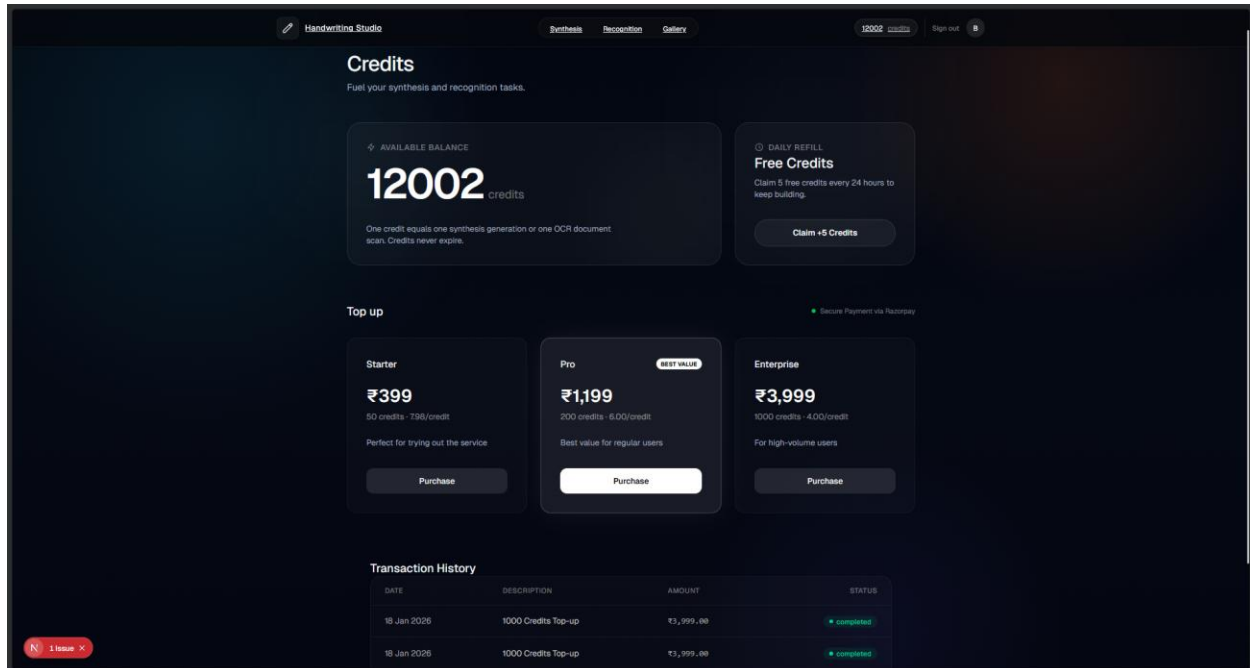
Search: - Use the search bar to find generations by text content - Filter by tags using the tag dropdown - Sort by date (newest/oldest)



5.3.6 Credits System

Checking Balance: - Your credit balance is displayed in the navigation bar - Each synthesis generation costs 1 credit - New users receive 10 free credits

Managing Credits: - Navigate to “Credits” or “Settings” page - View usage history - Purchase additional credits if needed



5.4 Test Cases Design

5.4.1 Authentication Test Cases

Test Case ID	Test Case Description	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC_001	Verify user registration with valid details	User does not exist	1) Open registration page. 2) Enter valid full name, email, password. 3) Submit form.	New user is created with unique userId; createdAt set to current time; credits = 10	As expected	Pass
TC_002	Verify registration fails for duplicate email	A user with same email exists	1) Open registration page. 2) Enter existing email + valid details. 3) Submit.	Registration rejected; "email already in use" error shown.	As expected	Pass
TC_003	Verify login with valid credentials	User account exists and verified	1) Open login. 2) Enter valid email & password (or Google OAuth). 3) Submit.	User session created; redirected to Dashboard.	As expected	Pass
TC_004	Verify login fails with wrong password	User exists	1) Open login. 2) Enter valid email + wrong password. 3) Submit.	Error shown; no session created.	As expected	Pass
TC_005	Verify Google OAuth login	User has Google account	1) Click "Continue with Google". 2)	User authenticated; account	As expected	Pass

			Complete OAuth flow.	linked if new.		
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5.4.2 Synthesis Generation Test Cases

Test Case ID	Test Case Description	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC_006	Generate handwriting with valid text	User logged in; credits > 0	1) Enter "Hello World". 2) Select style 9. 3) Click Generate.	SVG generated; displayed in preview; 1 credit deducted.	As expected	Pass
TC_007	Verify empty text rejection	User logged in	1) Leave text field empty. 2) Click Generate.	Validation error: "Text cannot be empty"	As expected	Pass
TC_008	Verify invalid character rejection	User logged in	1) Enter "Hello @World". 2) Click Generate.	Validation error listing invalid characters.	As expected	Pass
TC_009	Verify line limit enforcement	User logged in	1) Enter text with 25 lines. 2) Click Generate.	Validation error: "Maximum 20 lines allowed"	As expected	Pass
TC_010	Verify character limit per line	User logged in	1) Enter line with 80 characters. 2)	Validation error: "Line exceeds 75 characters"	As expected	Pass

			Click Generat e.			
TC_011	Verify insufficient credits error	User logged in; credits = 0	1) Enter valid text. 2) Click Generat e.	Error: “Insufficient credits”; generation blocked.	As expected	Pass
TC_012	Verify style range (0-12)	User logged in	1) Attempt style = 15 via API.	Validation error: style must be 0- 12	As expected	Pass
TC_013	Verify bias range (0.0-1.5)	User logged in	1) Set bias = 2.0. 2) Click Generat e.	Validation error: bias must be 0.0- 1.5	As expected	Pass

5.4.3 Gallery Test Cases

Test Case ID	Test Case Description	Pre- Conditions	Test Steps	Expected Result	Actual Result	Stat us
TC_014	Save generation to gallery	User logged in; generation exists	1) Generat e handwri ting. 2) Click “Save to Gallery” .	Generation saved with unique id; appears in gallery.	As expected	Pass
TC_015	View gallery items	User logged in; has saved items	1) Navigat e to Gallery tab.	All user’s generations displayed in grid view.	As expected	Pass
TC_016	Mark generation as favorite	User logged in; item exists	1) Click heart icon on item.	isFavorite set to true; icon filled.	As expected	Pass

TC_017	Add tags to generation	User logged in; item exists	1) Click tag icon. 2) Enter "important". 3) Save.	Tag added to generation's tags array.	As expected	Pass
TC_018	Delete generation	User logged in; item exists	1) Click trash icon. 2) Confirm deletion.	Generation removed from database; removed from view.	As expected	Pass
TC_019	Filter by favorites	User has favorites	1) Click "Favorites" filter.	Only items with isFavorite=true shown.	As expected	Pass
TC_020	Search generations by text	User has generations	1) Enter search term. 2) Press Enter.	Only matching generations displayed.	As expected	Pass

5.4.4 Realistic Rendering Test Cases

Test Case ID	Test Case Description	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC_021	Apply realistic rendering	User logged in; generation saved	1) Select generation. 2) Click "Make Realistic". 3) Select paper=cream, ink=ball point. 4) Apply.	Realistic PNG generated; realisticPng field populated.	As expected	Pass
TC_022	Verify paper type options	User on realistic modal	1) Open paper type	Options: white, cream, aged,	As expected	Pass

			dropdo wn.	lined, grid, recycled.		
TC_023	Verify ink type options	User on realistic modal	1) Open ink type dropdo wn.	Options: ballpoint, gel, fountain, marker, pencil.	As expected	Pass
TC_024	Verify wear level bounds	User on realistic modal	1) Adjust wear level slider.	Slider constrained to 0.0-1.0 range.	As expected	Pass
TC_025	Download realistic PNG	Realistic rendering complete	1) Click "Downl oad PNG".	PNG file downloaded with paper texture and ink effects.	As expected	Pass

5.4.5 Batch Processing Test Cases

Test Case ID	Test Case Description	Pre-Conditions	Test Steps	Expected Result	Actual Result	Stat us
TC_026	Create batch job	User logged in; credits >= variants	1) Enter text. 2) Select 3 styles. 3) Click "Genera te Batch".	BatchJob created with status=PEN DING; 3 generations queued.	As expected	Pass
TC_027	Verify batch progress tracking	Batch job processing	1) View batch jobs panel.	Progress bar updates; completedC ount increments.	As expected	Pass
TC_028	Verify batch completion	All variants generated	1) Wait for batch to complet e.	Status=COM PLETED; all generations in gallery.	As expected	Pass

TC_029	Verify insufficient credits for batch	User credits < variants	1) Select 5 styles. 2) Have only 3 credits.	Error: "Insufficient credits for batch"; job not created.	As expected	Pass
TC_030	Verify variant limit (1-13)	User logged in	1) Try to select 15 styles via API.	Validation error: max 13 variants allowed.	As expected	Pass

5.4.6 API Integration Test Cases

Test Case ID	Test Case Description	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC_031	Verify API health check	API server running	1) GET /health endpoint.	Response: status=healthy, model info returned.	As expected	Pass
TC_032	Verify unauthorized API access	No session token	1) Call protected endpoint without auth.	401 Unauthorized error returned.	As expected	Pass
TC_033	Verify API rate limiting	Multiple rapid requests	1) Send 100 requests in 10 seconds.	Rate limiting applied; 429 Too Many Requests after threshold.	As expected	Pass
TC_034	Verify CORS configuration	Request from allowed origin	1) Make request from frontend domain.	Request allowed; proper CORS headers returned.	As expected	Pass

TC_035	Verify file upload limits	User uploading file	1) Try to upload 10MB SVG file.	Error: file exceeds 4MB limit.	As expected	Pass
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5.4.7 Security Test Cases

Test Case ID	Test Case Description	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC_036	Verify ownership check on gallery	User A and User B exist	1) User A tries to access User B's generation by ID.	NOT_FOUND error returned (not FORBIDDEN).	As expected	Pass
TC_037	Verify password hashing	User registered	1) Check database for password field.	Password stored as bcrypt hash, not plaintext.	As expected	Pass
TC_038	Verify session expiration	Session exists	1) Wait for session to expire. 2) Try to access protected route.	Redirect to login; session invalidated.	As expected	Pass
TC_039	Verify XSS prevention	User logged in	1) Enter <code><script>alert('xss')</script></code> as text.	Text escaped/sanitized; no script execution.	As expected	Pass

TC_040	Verify SQL injection prevention	User logged in	1) Enter ' ; DROP TABLE users ; -- as text.	Query parameterized; no SQL injection.	As expected	Pass
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